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Stellar Populations of Star Clusters in the Local Group Dwarf Galaxy NGC 6822 Using Keck Spectroscopy

M.S. Project

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ABSTRACT

Star clusters have long served as records of the formation and evolutionary history of their host galaxies. While globular clusters (GCs) are well-studied, the origins of extended clusters (ECs) remain poorly understood. Using *Euclid* Early Release Observations, we identified seven star clusters in the halo and outskirts of the dwarf irregular galaxy NGC 6822 for follow-up spectroscopy with the Keck Cosmic Web Imager (KCWI). We derived radial velocities and stellar populations to investigate the host galaxy's history along with the nature of the clusters themselves. We find that the cluster kinematics show no strong correlation with previously proposed galaxy rotational models, suggesting no single previous large merger event. Our five ECs exhibit diverse stellar populations, including unexpectedly low metallicity in the youngest targets and two ECs with older populations than our GCs. Most notably, we detect significant internal rotation in the previously studied GC SC7, which is oriented orthogonally to the host galaxy's HI disk. We suggest SC7 as a possible stripped nucleus of an accreted satellite, providing evidence towards a hierarchical assembly of this dwarf galaxy.

Keywords: Local Group, dwarf galaxy, star clusters, extended clusters

1. INTRODUCTION

Tracing the formation history of galaxies is often achieved by studying their subcomponents in detail. Star clusters are important for this purpose since they can be long-lived and preserve information about kinematics and chemical abundances from different formation epochs. Among old cluster populations, globular clusters (GCs) are a widely used tracer, having typical masses of $\sim 10^4$–$10^5 M_\odot$ and half-light radii of $R_h \sim 2$–4 pc (see Brodie & Strader 2006). At high masses and large sizes ($M \gtrsim 10^6 M_\odot$ and $R_h \sim 10$–100 pc; Mieske et al. 2008; Brodie et al. 2011), ultracompact dwarfs (UCDs) blur the lines between dwarf galaxies and massive star clusters. Scenarios for UCD formation include tidal stripping of nucleated dwarfs, merging of star clusters, or fragmentation of cold streams at high redshifts (e.g., Norris et al. 2014; Fellhauer & Kroupa 2002; Chen et al. 2025).

A more mysterious type of star cluster, the extended clusters (ECs), has been found in the halos of both giant and dwarf galaxies (Huxor et al. 2005; Hwang et al. 2011). ECs are larger than GCs, with $R_h \sim 10$–30 pc (Hwang et al. 2011; Tanvir et al. 2012) and masses of $\sim 10^4$–$10^5 M_\odot$. Diffuse and "faint fuzzy" (FF) star clusters are similar to the ECs, but associated with galaxy disks (Larsen & Brodie 2000; Brodie & Larsen 2002; Peng et al. 2006). It is not known if there is a clear boundary between ECs/FFs and UCDs (Forbes et al. 2013). Proposed scenarios for EC/FF origins include remnants of tidally stripped galaxies, collisions of star clusters, variations in birth sizes tied to cloud pressure, triggered formation from galaxy interactions, and black-hole driven evaporation (Elmegreen 2008; Hurley & Mackey 2010; Hwang et al. 2011, 2014; Gieles et al. 2021; Romanowsky et al. 2023).

Star clusters are also deeply connected to studies of galaxy halos. Since the Searle & Zinn (1978) study of GC metallicity distributions in the Milky Way (MW) halo, galaxies have been understood as having hierarchical assembly histories. In simulations of low-mass galaxies, it was found that the outskirts of these galaxies tend to be more metal-poor than areas closer to the center, and that halos are likely to experience a major and several minor merger events in their lifetime (Tau et al. 2025; Deason et al. 2022). Low mass galaxies also seem to have halos of more than 90 percent in-situ stars when looking at galaxies that have had merger events (Cooper et al. 2025). In standard theories for the formation of GC systems, the contributions from accreted galaxies are strongly correlated with host galaxy

mass, so that GCs in dwarf galaxies are almost exclusively formed in-situ (Choksi & Gnedin 2019). Therefore, GCs in dwarfs should provide relatively pristine records of their formation without the complicating effects of mergers.

The *Euclid* space telescope has begun to provide wide-field, high-resolution images of nearby dwarf galaxies, studied in greater detail than before (Hunt et al. 2025). In particular, their stellar halos and star clusters can be resolved. Following up on their clusters with spectroscopy provides additional, fundamental information about their kinematics and stellar populations.

Here we examine the nearby galaxy NGC 6822, which is a dwarf Irregular (dIrr) in the Local Group, at a distance of 508 kpc (Fusco et al. 2012), a stellar mass of 10^8 $M_\odot$ (Nersesian et al. 2019), and half-light radius of ∼400 pc (McConnachie 2012). From proper motion confirmation with Gaia, it was shown that this galaxy is the nearest star-forming dwarf that has not yet interacted with the MW or M31, so that NGC 6822 has been and continues to be a prime target for observations (e.g., McConnachie et al. 2021; Dalcanton et al. 2024; Chown et al. 2025; Koch et al. 2025). One of the predominant features of this galaxy is an extended HI disk that rotates off-axis compared to an older spheroidal stellar structure (de Blok & Walter 2000, 2006; Demers et al. 2006), which shows a more dense region in the northwest that may be a small companion dwarf galaxy, as well as a large "hole" in the southeastern part. Hunt et al. (2025) showed that the young star distribution overlaps with the HI disk. Different proposed models in order to explain this appear in Veljanoski et al. (2015), which takes a look at whether or not clusters belonging to the system support a major merger event in the past.

In this paper, we analyze spectra obtained from the Keck telescope for seven of these star clusters around the NGC 6822 (Section 2). We obtain radial velocity measurements as well as information about the clusters' stellar populations (Section 3). We discuss our results in Section 4. More information on our methods can be found in Appendix A.1.

2. METHODS

2.1. *Target selection*

Following the release of the Euclid ERO images of NGC 6822 described in Hunt et al. (2025), but prior to the paper on star clusters by (Howell et al. 2026, hereafter H+26), we carried out a visual search for "new" star clusters around NGC 6822, using the previously known star clusters as a guide to their appearance (Veljanoski et al. 2015). We inspected the Euclid imaging using the ESASky viewer[1], and Subaru Hyper Suprime-Cam (HSC; Miyazaki et al. 2018) *gri* color imaging using the hscMap viewer[2]. We readily identified the previously known clusters outside the main body of the galaxy and within the Euclid field of view (SC3, SC6, SC7), We also discovered six new clusters: SC8, SC9, ZC1, ZC2a, ZC2b, and ZC3. The latter were independently discovered by H+26 with the identifications ESCC-27, ESCC-33, ESCC-23, ESCC-28, ESCC-29, and ESCC-22. ZC2a and ZC2b are so named because we initially thought they might be two sub-clumps belonging to one cluster. However, we later determined them to be independent based on differences in radial velocity, and treated them as such. ZC1, ZC3, and SC9 all have similar blue colors, compact sizes, and radial distances just beyond the visible edge of the galaxy. SC8 is the largest and faintest of the new clusters – a remarkable new EC – and also the farthest from the center.

We chose seven clusters for spectroscopy, incluing three previously studied (SC3, SC6, SC7; Hwang et al. 2014; Veljanoski et al. 2015) and four newly identified (SC8, SC9, ZC2a, and ZC2b). We also obtained spectra for SC5, but the S/N was too low for meaningful measurements. The left panel of Figure 1 shows the locations of the target clusters around their host galaxy, superimposed on the Euclid image. The right panel shows zoomed-in images of these clusters, with Euclid on the left, and HSC on the right. Using the HSC images, the blue colors of SC9 and especially ZC2a,b are visible, while the other clusters are much redder. SC6 and SC7 are symmetrical and very dense, with SC7 showing elongation. SC3, and SC8 in particular, are diffuse.

Basic properties of the targets are summarized in Table 1. The sizes and absolute magnitudes are shown in Figure 2, along with the rest of the star cluster sample from NGC 6822 (H+26). Note that the sizes are likely systematically underestimated by up to a factor of 2, particularly for the ECs (see figures 8 and 9 from H+26). The clusters are color-coded by their ages, inferred from spectral energy distribution (SED) fitting in H+26. GCs SC6 and SC7 are part of the dense GC population while other ECs fall into the sparse part on the left side. Both the very youngest and oldest objects all appear to be within 5 half-light radii, with younger objects appearing fainter in general while older objects can range from $M_V \sim -3$ to -9 mag.

[1] https://sky.esa.int/esasky/

[2] https://hscla.mtk.nao.ac.jp/hscMap

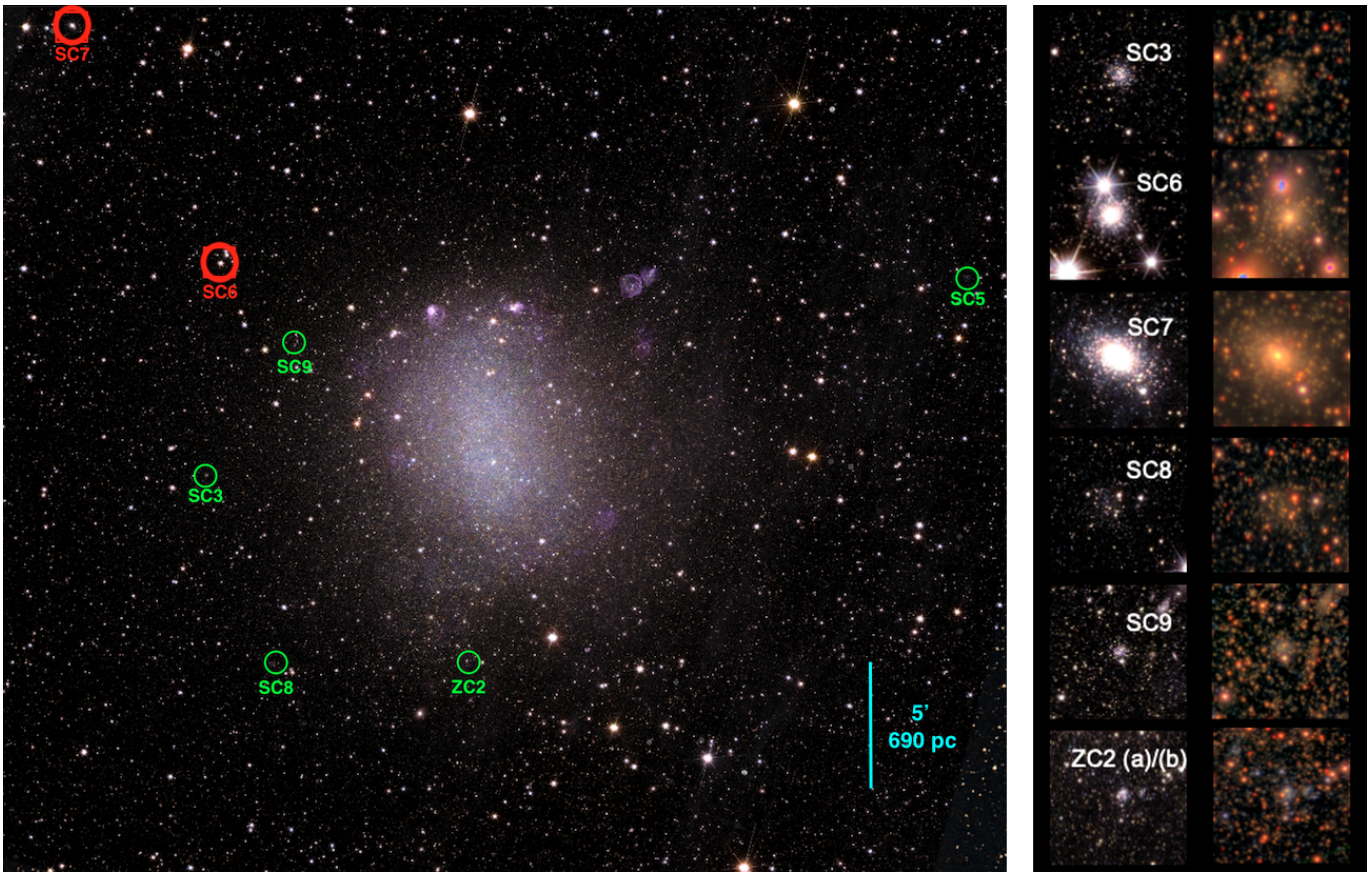


Figure 1. Imaging of the dwarf galaxy NGC 6822 and its star clusters that were targeted with KCWI. Left panel shows a wide-view (with scale-bar included) using the Euclid VIS I_E filter as well as the NISP Y_E and H_E filters (ESAskyviewer; Giordano et al. 2018). Symbols mark star clusters studied here: GCs in red and ECs in green. Other previously measured clusters SC1, SC2, and SC4 all lie along the axis of rotation of the HI disk. North is up and East is left. The right panel includes our clusters captured with *Euclid* in the left column, and HSC in the right column. The field of view of each thumbnail is 50 × 30 arcsec (∼ 120 × 70 pc).

ID	*Euclid* ID	RA (J2000)	Dec (J2000)	V [mag]	R_h [pc]	Distance (kpc)	Exposures (s)
SC3	ESCC-05	19:45:40.1	−14:49:25.3	18.06	7.15	1.53	3 × 450
SC6	ESCC-07	19:45:36.9	−14:41:11.7	15.77	2.7	1.74	1 × 900
SC7	ESCC-08	19:46:00.8	−14:32:35.7	15.26	2.3	3.23	2 × 300, 90, 140
SC8	ESCC-27	19:45:29.9	−14:56:28.55	17.92	12.41	1.68	3 × 600
SC9	ESCC-33	19:45:27.5	−14:44:50.6	17.941	3.44	1.17	600, 900
ZC2(a)/(b)	ESCC-28/29	19:44:59.8	−14:56:24.6	17.92/17.1	12.41/4.62	1.21	3 × 600

Table 1. Spectroscopic targets, with basic properties from the literature. The number of exposures, and total exposure time for our spectroscopy. *Euclid* ID cluster names, RA, Dec, V mag R_h, and distance from the center of the galaxy from H+26.

2.2. *Follow-up spectroscopy*

We next carried out spectroscopic observations using the Keck Cosmic Web Imager (KCWI; Morrissey et al. 2018). The observations were made over several nights on October 10, 30, and November 8 2025, as well as April 23 2026 (all dates in HST). Both weather and seeing conditions varied. The setup used the Medium slicer, with a slice width of 0.69 arcsec and a field of view of 16.5 × 20.4 arcsec. The blue side used the BL grating, and the red side used either RL or RH2, but for this work we analyze only the blue data. These spectra have a resolution of $R = 1800$ and wavelength

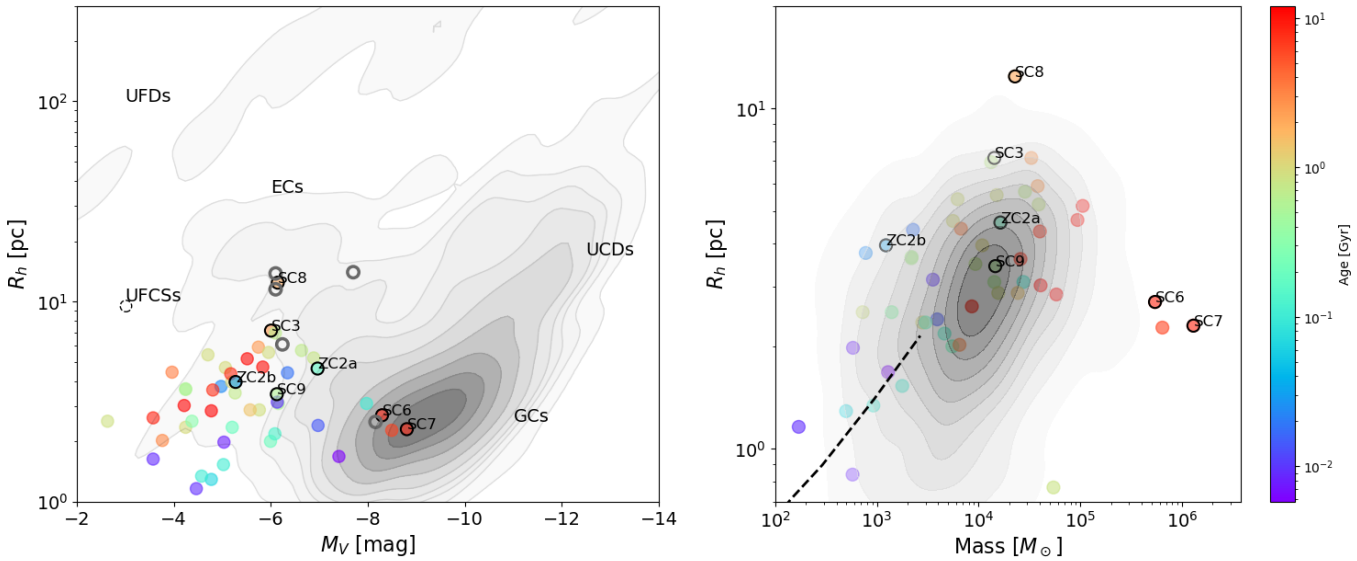


Figure 2. Relationship between the size and absolute magnitude of star clusters (from Brodie et al. 2011 and H+26). Labels in left panel mark approximate regions for UCDs, GCs, and ECs. Regions for ultra-faint dwarfs (UFDs) and ultra-faint compact satellites (UFCSs) are also indicated (Simon 2019; Cerny et al. 2026). Right panel shows the NGC 6822 clusters. Points are color-coded by age inferred from SED fitting (H+26), except for open grey circles for ECs beyond the Euclid field of view (Hwang et al. 2014). Spectroscopic targets from this work are labeled and marked with dark outlines.

range of 3500–5600 Å. The number of exposures and exposure times for each target are given in Table 1. We observed SC3, SC6, SC7, SC8, SC9, ZC2a, and ZC2b (the latter two clusters within the same pointing).

2.3. *Data Preparation*

We reduced the spectra with `kcwi drp`[3] (Neill et al. 2023), with sky subtraction disabled so we could carry out our own manual sky subtraction. The heliocentric corrections were also disabled, and we calculated these using `astropy`. We used the mosaicing package `MONTAGE`[4] to resize the stretched images as well as to stack each exposure. We then used `QFitsView`[5] to extract spectra from the datacubes. For sky subtraction, we used the same number of pixels in each column as in the object extraction, choosing a sky region that was directly above in these same columns. This procedure avoids any effect from a residual gradient in the KCWI data (see Appendix A of Gannon et al. 2020). For the GCs SC6 and SC7, we used aperture sizes that gave us the highest SNR. Other clusters had considerations of contaminant stars to take into account, where aperture shape would vary.

Given these large apertures sizes and the low Galactic latitude of −18 deg, there is potential contamination from foreground stars, which must be distinguished from stars belonging to the clusters. In particular, there are relatively bright red stars overlapping SC8, SC9, and ZC2 (see Figure 1). We used three methods to exclude MW stars. First, we constructed color–magnitude diagrams using Euclid photometry and determined which bright stars were consistent or not with the trends for stars in NGC 6822 (Hunt et al. 2025 figures 9 and 11) – see more details in Appendix A.1. Second, we crossmatched stars with search radius of just the cluster as well as 90 sec arcsecond radius with the Gaia catalogue to look for any MW foreground stars using their proper motion. Third, we examined the gravitationally-sensitive lines between 510 and 520 nm, and looking for how deep they are when looking at spectra of only the possible contaminant stars. With deeper, more prominent lines it helps add evidence towards these stars not being not part of the cluster, and instead being a foreground star. An RGB star that is truly part of the cluster would have a smaller gravitational effect on our absorption lines than a main-sequence foreground star.

2.4. *Spectral analysis*

[3] https://kcwi-drp.readthedocs.io/en/latest/

[4] http://montage.ipac.caltech.edu

[5] https://www.mpe.mpg.de/~ott/QFitsView/

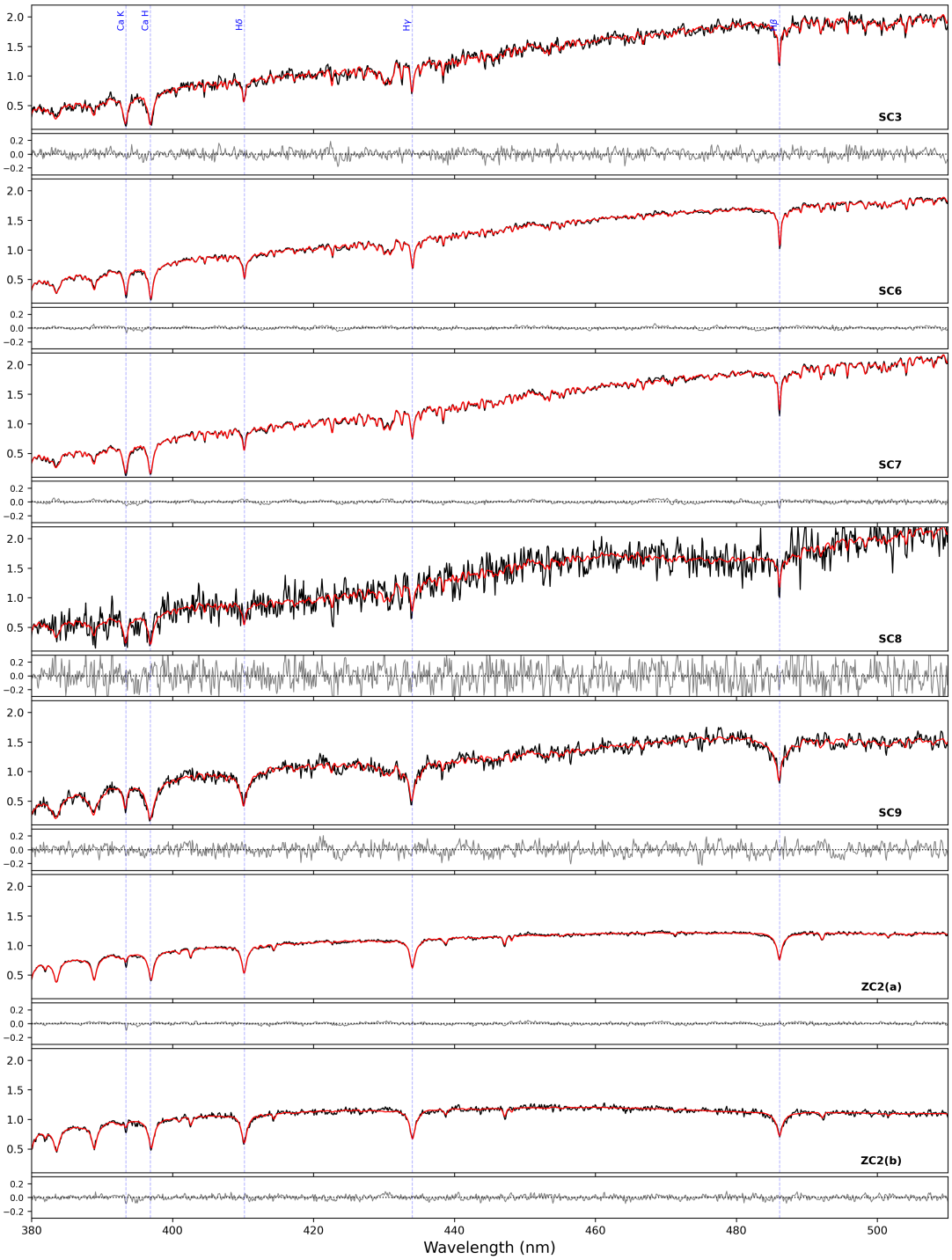


Figure 3. KCWI spectra from the clusters (black) along with their pPXF fits (red).

Once we had extracted spectra from each stacked datacube, the next step was to use the Penalized Pixel-Fitting (pPXF) package (Cappellari & Emsellem 2004) to extract quantities such as radial velocity, age, and metallicity of an object. For the stellar template library, we adopted E-MILES (Vazdekis et al. 2016), with a spectral range of 1680 to 50000 Å. This template covers an age range of 0.03 to 14 Gyr, and a metallicity range from −2.27 to +0.40 dex. To validate our methods, we analyzed archival spectra of dwarf galaxy GCs with previously known stellar population values, and found our values to be consistent with the literature (see details in Appendix A.1).

We fitted each spectrum in two passes. First, we fitted for radial velocity, with results that were robust to starting guesses and the degrees of the additive and multiplicative polynomials used in modeling the continuum. We then ran pPXF a second time with fixed velocity, performing a grid search over a wide range of polynomial degrees, and adopted the median values obtained for mass-weighted age and metallicity. Using a multiplicative degree in the polynomial also assists in reducing extinction from the MW. Figure 3 shows each of our spectra. Even with a high exposure time relative to other clusters, SC8 has the lowest SNR due to the extended structure (low surface brightness) of the object. For both ZC2 and SC9, our priors assumed the cluster is between 0–2 Gyr in age in our spectral fit, due to the strong blue color that the cluster appears as in HSC imaging. These clusters are sensitive to initial conditions, where solutions of old ages can be chosen in both our polynomial degree grid search and our bootstrap method for errors.

Instead of using the pPXF built-in function for radial velocity errors (equations (5c) and (5d) in Cappellari (2023)), we instead used the bootstrapping method shown in the pPXF github[6], while fixing the polynomial degrees from our grid search. We also used this method to derive the uncertainties in age and metallicity. We used the difference in the 84th and 16th percentiles of our distributions obtained from bootstrapping. For cases with zero uncertainty from bootstrapping, we used the difference between the principal value and the closest value in the template grid, divided by 2. We then used the built-in function of pPXF to calculate the mass-to-light ratios in the V-band.

3. RESULTS

3.1. *Radial Velocities*

All results can be seen in Table 2. Comparing to previous spectroscopic results in the literature, the radial velocities of SC6 and SC7 are consistent with the literature values while SC3 differs (see Figure 13). For ZC2(a) and ZC2(b), the velocity difference supports these as two distinct clusters (-36 and -24 km s^{-1}) at present day. The other clusters all lie within the velocity range of previously studied clusters of this galaxy.

ID	R_V (km s^{-1})	Literature R_V (km s^{-1})	Metallicity (M/H)	Age (Gyr)	M/L_V	S/N (per Å)
SC3	-45 ±4	-83 ±14	-0.96	8.5	3.34	29.9
SC6	-3 ±2	-6 ±3	-1.49	6.5	3.08	93.6
SC7	-27 ±1	-37 ±2	-1.26	8	2.94	83.3
SC8	-37 ±8	-	-0.96	9	2.88	12.8
SC9	-34 ±6	-	-0.66	0.8	.83	24.5
ZC2(a)	-36 ±2	-	-2.27	0.08	.21	107.1
ZC2(b)	-24 ±2	-	-2.27	0.06	.23	54.3

Table 2. The results from KCWI spectroscopy, including the radial velocity (R_V) for each cluster, the previous R_V value found in Veljanoski et al. (2015), metallicity, age, stellar mass-to-light ratio M/L_V, and S/N.

Given an expanded dataset of star cluster velocities, we test if they fall in line with either the cigar or disc model history, or neither. The cigar model correlates to an axis of rotation shared with the HI disc, while the disc model would be an axis perpendicular to it. As shown in Figure 4 (right), our measurements more closely align with the disk model, although there is no strong correlation towards one model or the other. This corresponds to results found in Hwang et al. (2014) of no preferred model.

One interesting investigation we did was to look for a net rotation in SC7 itself, due to its elongated shape when compared to a classical GC (see shape in Figure 7). We took spectra from either side of it as seen in figure 11, and found a difference in velocity. Our choice in wedge shape is from Cappellari (2016), while using the Integrated Field Unit (IFU) of the KCWI. The difference between the left and right sides of this cluster came to a difference of 14 km s^{-1}. This is comparable with several GCs studied in Petralia et al. (2024). We found a similar rotational difference in SC6, even though it appears to be more spherical in shape.

3.2. *Stellar Populations*

We find that the youngest clusters, ZC2(a)/(b), have the same [M/H] of -2.27 dex, which is a surprisingly low metallicity for a young cluster. SC9 is the second youngest cluster but we measure its metallicity to be in the -0.4

[6] https://github.com/micappe/ppxf_examples/blob/main/ppxf_example_population_bootstrap.ipynb

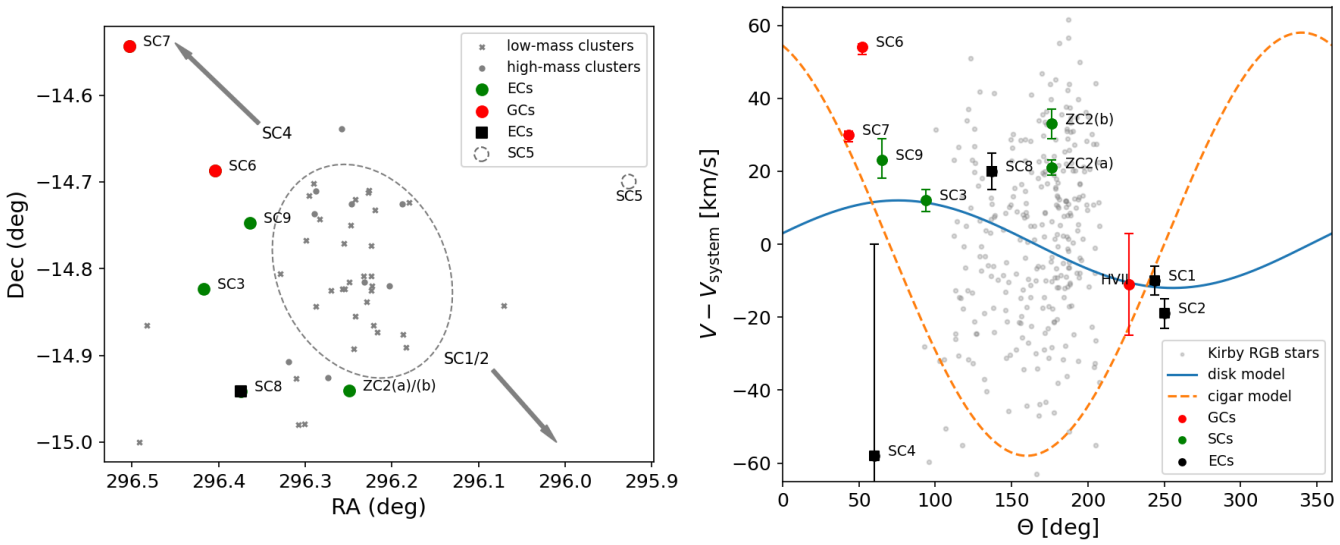


Figure 4. Left panel shows the distribution of clusters. Low and high mass clusters from H+26. Grey-dashed line is the r_{25} size of the galaxy from McConnachie (2012) Green arrows show locations of clusters further away. Right panel shows how closely the observed velocities of the clusters correlate to either rotation model based on their position angles around the galaxy. Disk model, cigar model, and literature objects (HVII, SC1, SC2, SC4) from Veljanoski et al. (2015). RGB stars along one axis of the galaxy taken from Kirby et al. (2013).

dex range. The previously measured globular clusters of SC6 and SC7 are about −1.1 and −1.7 dex, which closely follow results found in Larsen et al. (2022), however appear younger than their age results of 13 Gyr. We find that clusters SC3 and SC8 appear to be the oldest clusters of our sample.

Stellar populations can also be seen in Figure 5 where the positions can be seen when compared to previous literature trends, including simulated (E-MOSAICS, Pfeffer et al. 2018) and observed (color–magnitude diagram) objects. Because H+26 took data of all clusters included in this paper, we their photometric data as reference points. We chose a cutoff of low mass of clusters below their measured value of $10^4 M_\odot$. In Figure 6 we can look for trends in how age and metallicity change radially outward from the center of the galaxy. Previous measurements made in Hwang et al. (2011) show that clusters SC1, SC2, SC3, and SC4 get both larger and bluer as you go out from the center. Our new clusters do not follow this trend, especially when looking at our GCs.

4. DISCUSSION/CONCLUSION

We were able to successfully fit our clusters to spectra and obtained radial velocities and stellar population information. The nature of these clusters have interesting quantities. Our two GCs are not so standard, while our other ECs differ in shape and population from one another. Both GCs have a rotation associated with them which is not common in GCs. We also suspect a nonhomogenous population of stars in the system radially outwards from the center, but will require further investigation. This raises questions on where both of these clusters originated from, and points to the possibility of SC7 possibly being a stripped nucleus, a previous dwarf galaxy that has had its halo stripped away (and hence might be classified as a UCD). Its structure can be seen in Figure 7. Theoretical work would indicate that in a galaxy at this mass, both of these GCs likely would have formed in-situ Choksi & Gnedin (2019). Georgiev et al. (2009) reports on GCs found in other dIrr galaxies that are suspected to be previously accreted dwarf galaxies due to their similarities in color and magnitude to M54 and ωCen. Their objects are close to the center of galaxies, while both SC6 and SC7 are further out into the halo. Overall, both GCs fall younger and more metal-poor than the trendline given by the E-MOSAICS work, which is a trend from Milky Way clusters.

One interesting counter to the theoretical work of in-situ clusters is the fact that the rotation of SC7 is orthogonal to the HI disk of the galaxy. If formed ex-situ at some time in the past, this would explain why the cluster does not follow the rotational behavior of the galaxy itself. The structural complexity along with the rotational difference points is also evidence towards this. This leads to questions if this cluster could have been accreted without an overall larger accretion event, or if it it was part of a single large one.

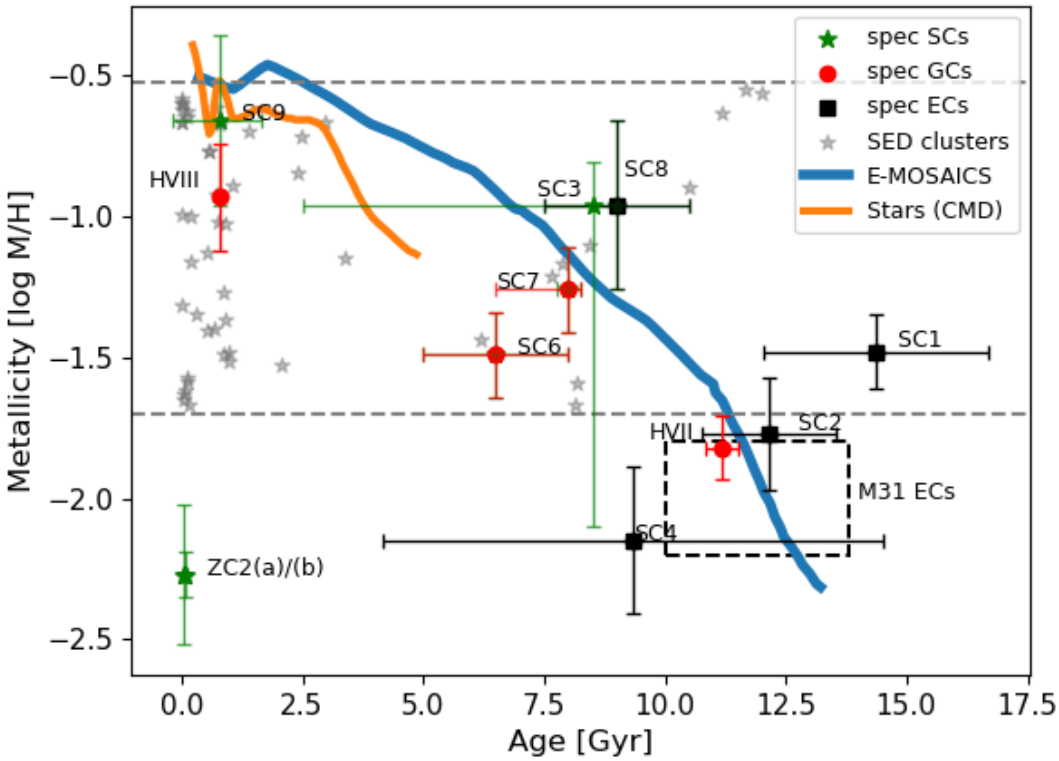


Figure 5. Our spectroscopic age–metallicity results compared to the full sample using SED fitting includng the grey dotted lines representing their priors in H+26. Green points represent cluster values from Hwang et al. (2014). An age–metallicity relation of GCs is also shown using E-MOSAICS (Horta et al. 2021) as well as one for stars in this same galaxy (Fusco et al. 2012). Black dashed line range of M31 ECs comes from Mackey et al. (2006).

The biggest question is the formation of ECs and how they differ to GCs. Hwang et al. (2014) posited that they may be stripped cores of dwarf galaxies or the products of cluster mergers, and indications that they may be survivors of tidal disruption events. Our youngest clusters, SC9 and ZC2 (a)/(b), interestingly have very different metallicity content. SC9 lies along either trend line with its error, while ZC2 is very far off in its own population. These both have foreground red stars that were excluded from the spectra extraction. Although young, SC9 is still on the order of 10 times older than either ZC2 cluster. SC3 and SC8 have very similar stellar populations to one another even though their shapes are very different from one another. The old ages of SC3 and SC8 show that extended clusters are able to retain their structure for a long period of time, where an open cluster could be more prone to disruption. Their large R_h of 7-12 pc would follow the faint-fuzzy description given to clusters SC3 and SC4 of this galaxy (Hwang et al. 2011), as well as a different SC1 cluster of the galaxy NGC 247 (Romanowsky et al. 2023). These two are also of interest because they appear to be older than either of the two GCs, which are classically thought to be some of the oldest objects in the universe.

Because our cluster velocities do not follow either proposed cigar or disk model, this is evidence against a large-scale merger event some time in this galaxy's past. If it closely followed the cigar model it would have indicated an accretion type event. Future analyses on more clusters at different position angles around the galaxy could bring this into question, because we are lacking any targets in a relatively large gap in position angle.

Current HSC imaging provides a preview of the depth and image quality of Rubin data. Both Rubin and eventually the Roman space telescope will observe other Local Group galaxies, which allows for NGC 6822 to be a useful first target to perform this analysis on, in order to prepare for datasets from these two new telescopes.

5. ACKNOWLEDGEMENTS

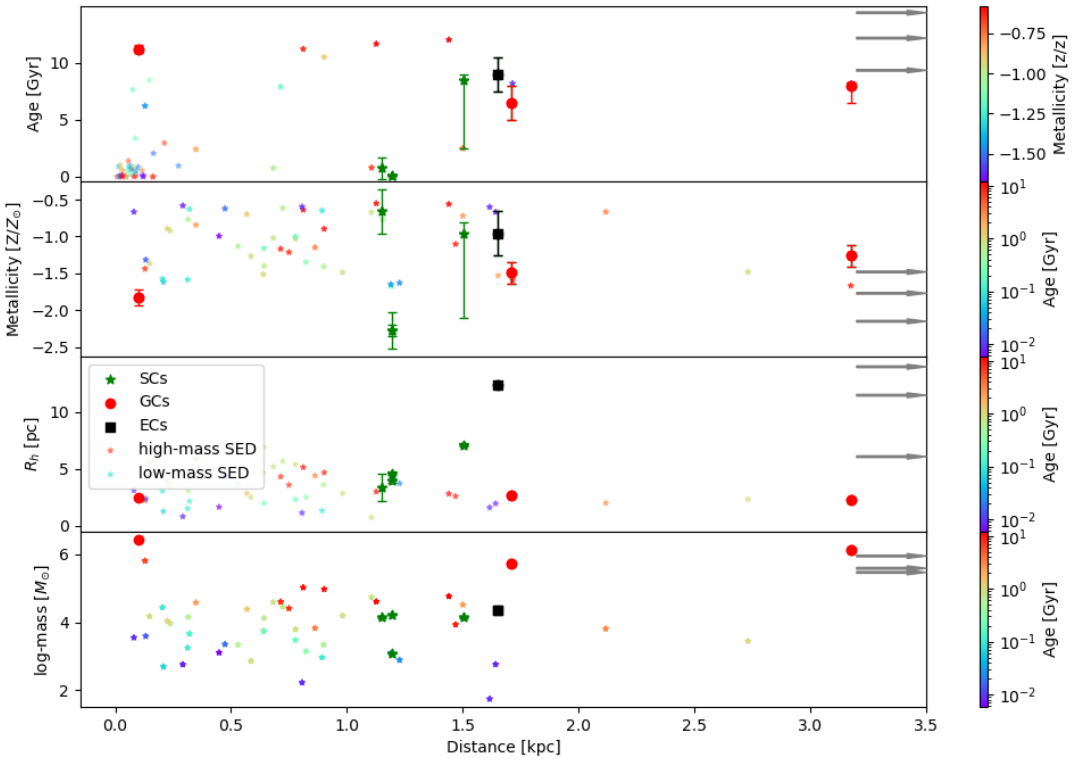


Figure 6. Age and metallicity variation with distance from the center of the galaxy. R_h as well as circle and x markers are from H+26, representing low-mass and high-mass objects, respectively. Green points represent Hwang et al. (2014) cluster values for HVII, SC1, SC2, and SC4. Green arrows indicating that clusters SC1, SC2, and SC4 go much further out, to distances of 10.7, 4.1, and 6.0 kpc, respectively.

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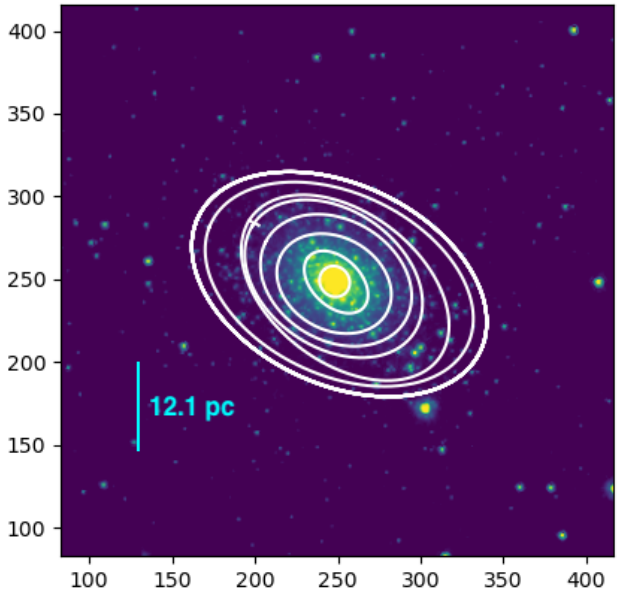


Figure 7. Isophotal analysis of SC7, showing the twisting as we go further from the center. Each pixel is 0.1 arcsec.

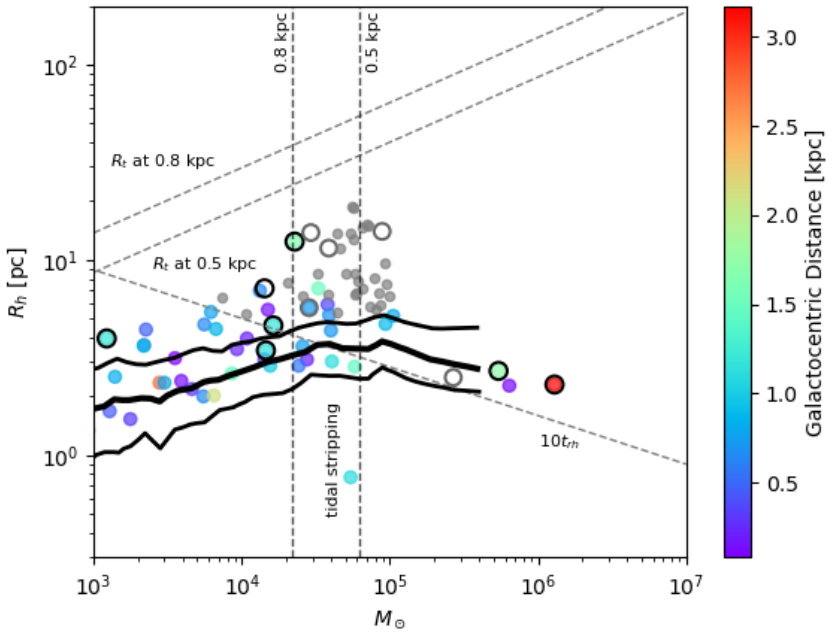


Figure 8. R_h plotted against $M_\odot$ values from H+26. A trend for the tidal radius for GCs at a galactocentric distance of 0.8 kpc given as a dashed line from Georgiev et al. (2009).

nology Office, Computation Technologies Project, under Cooperative Agreement Number NCC5-626 between NASA and the California Institute of Technology.

Software: astropy (Astropy Collaboration et al. 2013, 2018) pPXF, matplotlib, Qfitsviewer,ds9,KOA, hscviewer, ESAsky, montage, kcwi drp

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A. APPENDIX

A.1. *Confirmation of Methods*

To confirm our spectroscopic analysis methods, we took GC spectra from the WiFeS Atlas of Galactic Globular cluster Spectra (WAGGS) project (Usher et al. 2017). Their data are available for download[7], and we combined the U and B gratings into one spectrum to mimic our KCWI spectra wavelength range. We chose objects at various age and metallicity values, picking objects from the SMC and the Fornax dwarf spheroidal. We selected a young object ($\ll 1$ Gyr) and several older objects (6–14 Gyr). Our objects include NGC 330, NGC 361, NGC 0121, Fornax 3 (NGC 1049), and Fornax 4. Figure 9 shows how our values compare.

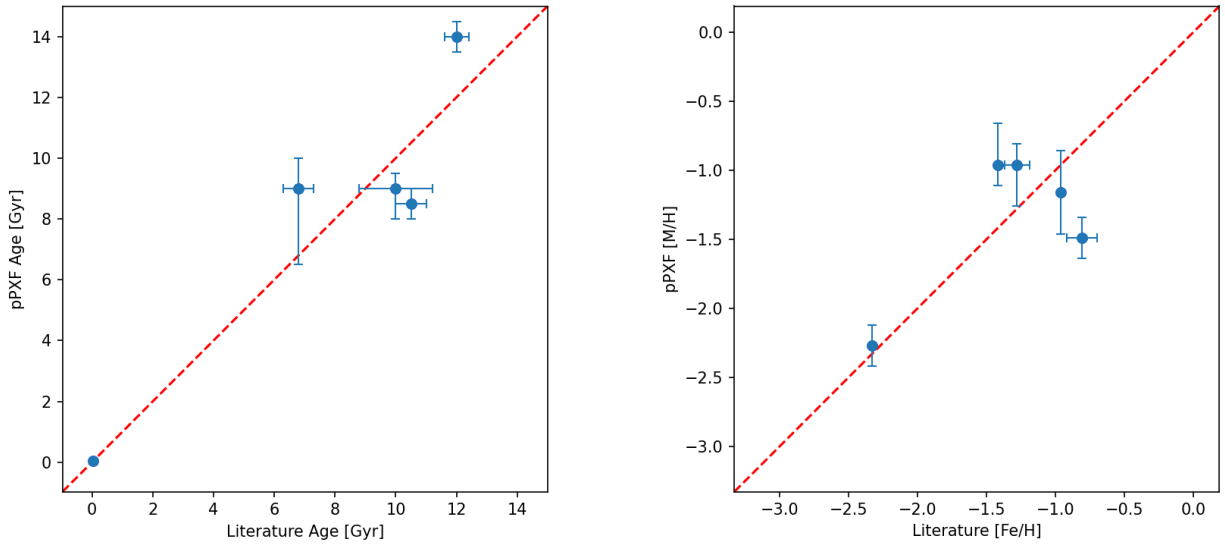


Figure 9. Age and metallicity results from our pPXF modeling of WAGGS spectra of GCs, compared to literature values (Table 1 from Usher et al. 2017).

Excluding foreground stars with Gaia was done based off of looking at color–magnitude diagrams of stars in the cluster and surrounding area as seen in Figure 10. This process was necessary for SC8, SC9, and ZC2 where we needed to decide which bright, redder stars could be excluded from our spectra extraction. Stars with good measurements of a relevant proper motion were excluded.

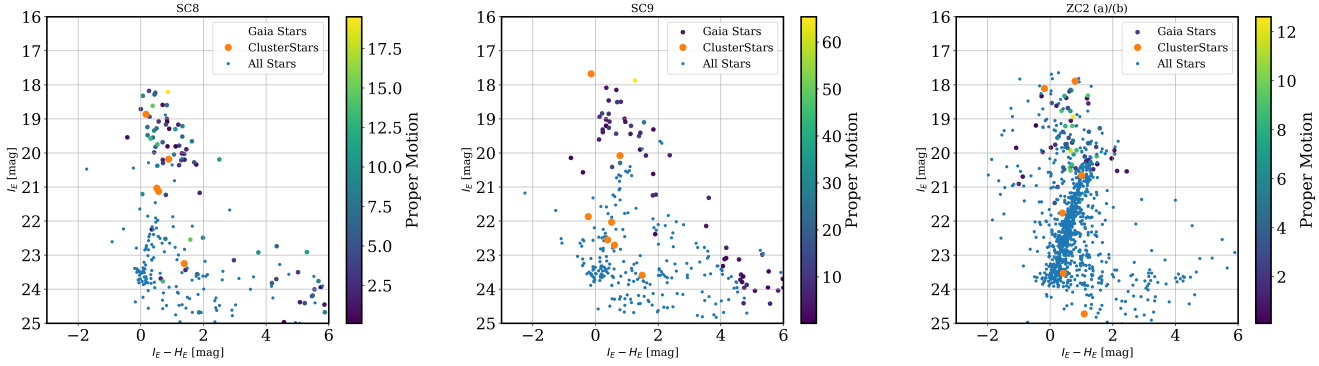


Figure 10. A color-magnitude diagram with Gaia objects colored by proper motion.

[7] https://cloud.datacentral.org.au/teamdata/WAGGS/public/

A.2. *Extra Figures*

Our pixel selection wedge shown in Figure 11. Figure 12 includes HST imaging of the ZC2 clusters, although analysis is out of the scope of this paper. Figure 13 shows how our results compare with previous literature values. Ages from Larsen et al. (2022) were assumed, not measured.

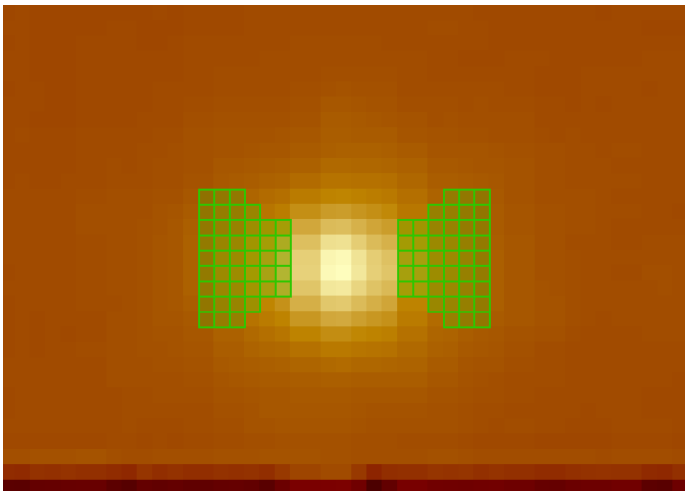


Figure 11. Our pixel selection for SC7 using QFitsView in order to find a radial velocity difference.

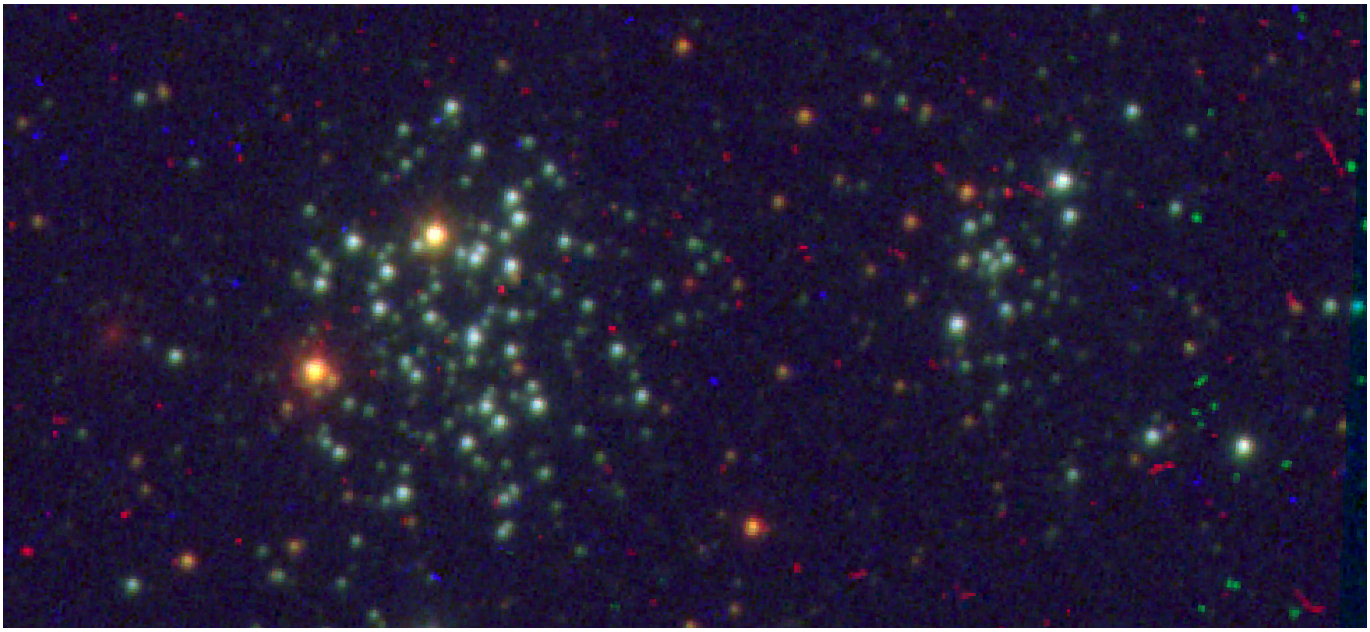


Figure 12. HST imaging of ZC2 (a)/(b), using WFC3 (F336W) and ACS (F475W, F814W).

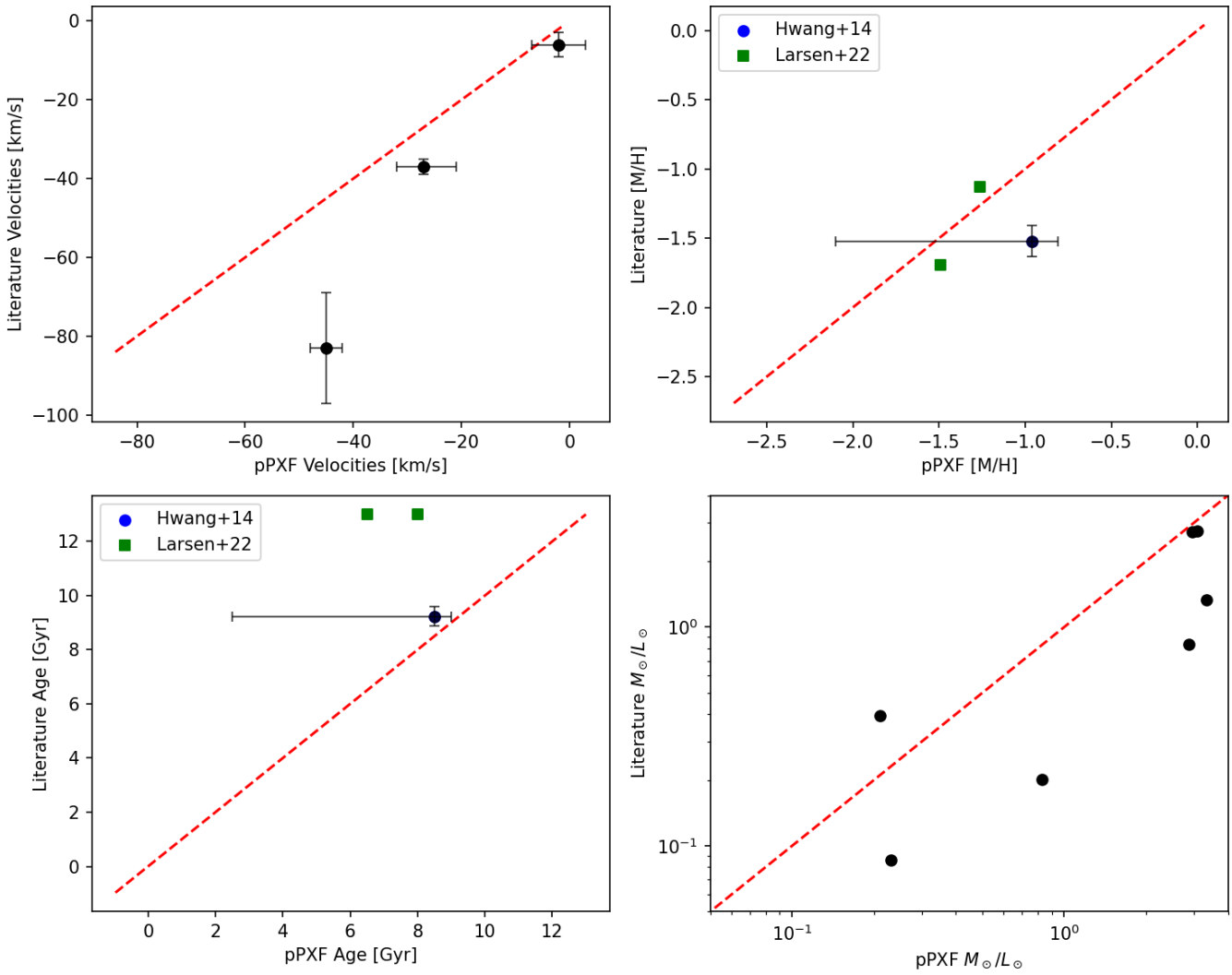


Figure 13. The values of this work compared to previous literature. Velocities from Veljanoski et al. (2015). Metallicity and Ages from Hwang et al. (2014); Larsen et al. (2022). $M_\odot/L_\odot$ values from H+26.

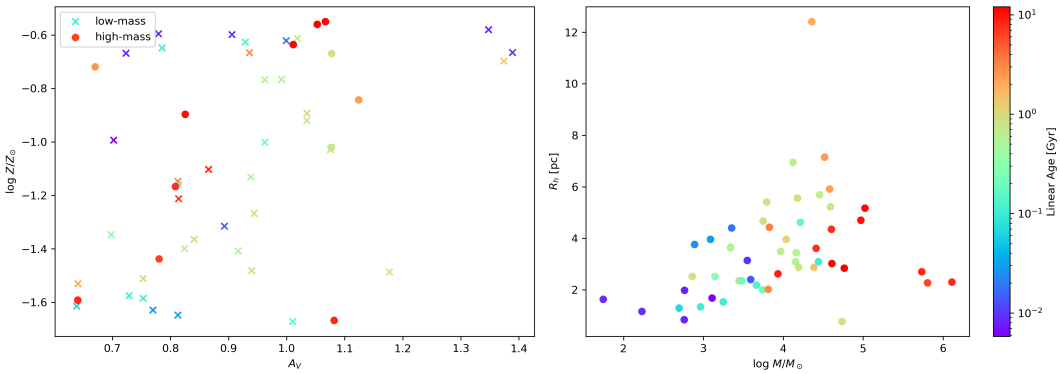


Figure 14. H+26 values shown for comparison. Left panel is Metallicity of clusters related to their A_V value. Right panel shows R_h related to their mass. Both are colored according to age.